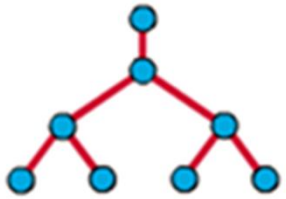
Bus Topology

Ring Topology

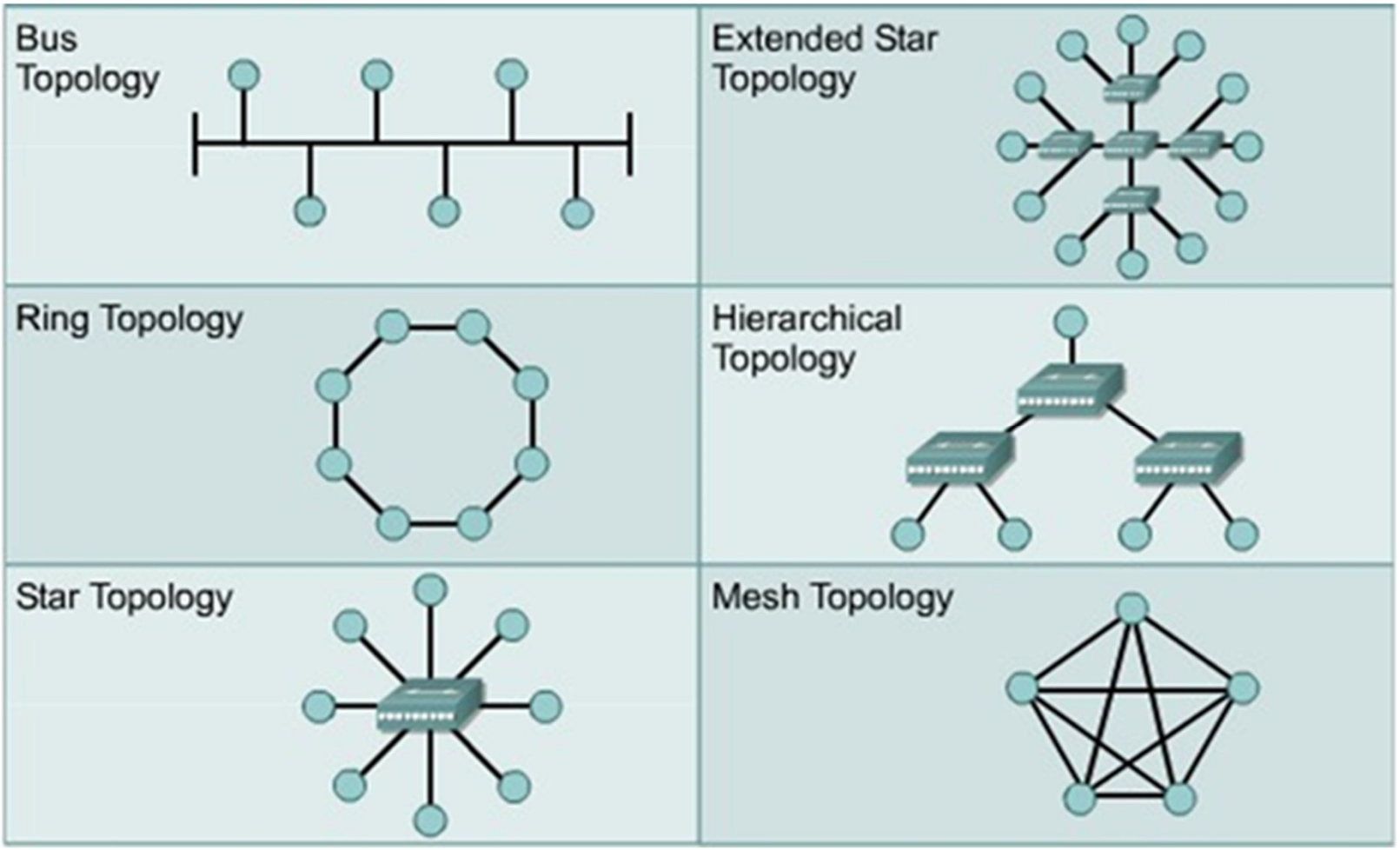
Star Topology

Extended Star Topology



Hierarchical Topology

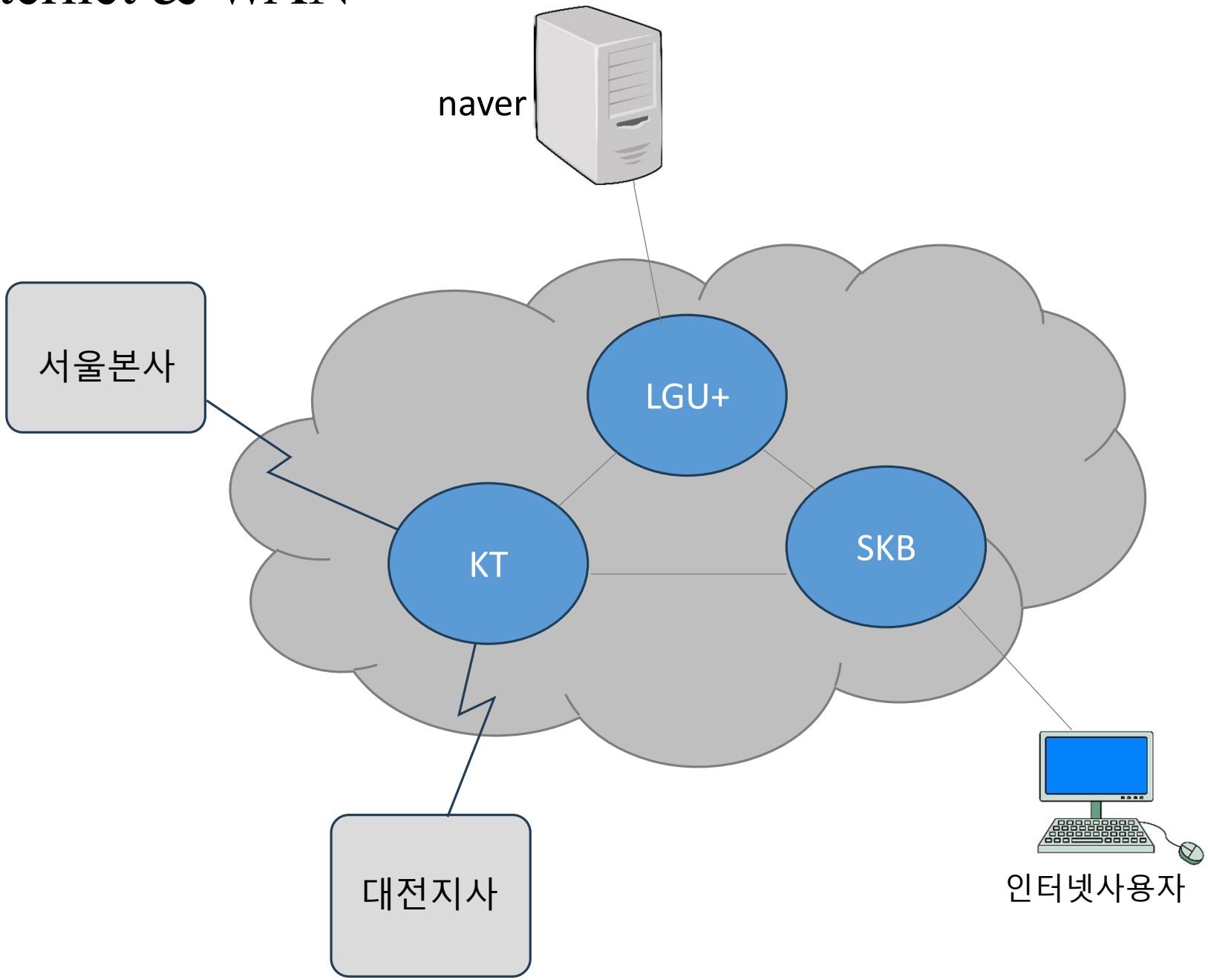
Mesh Topology



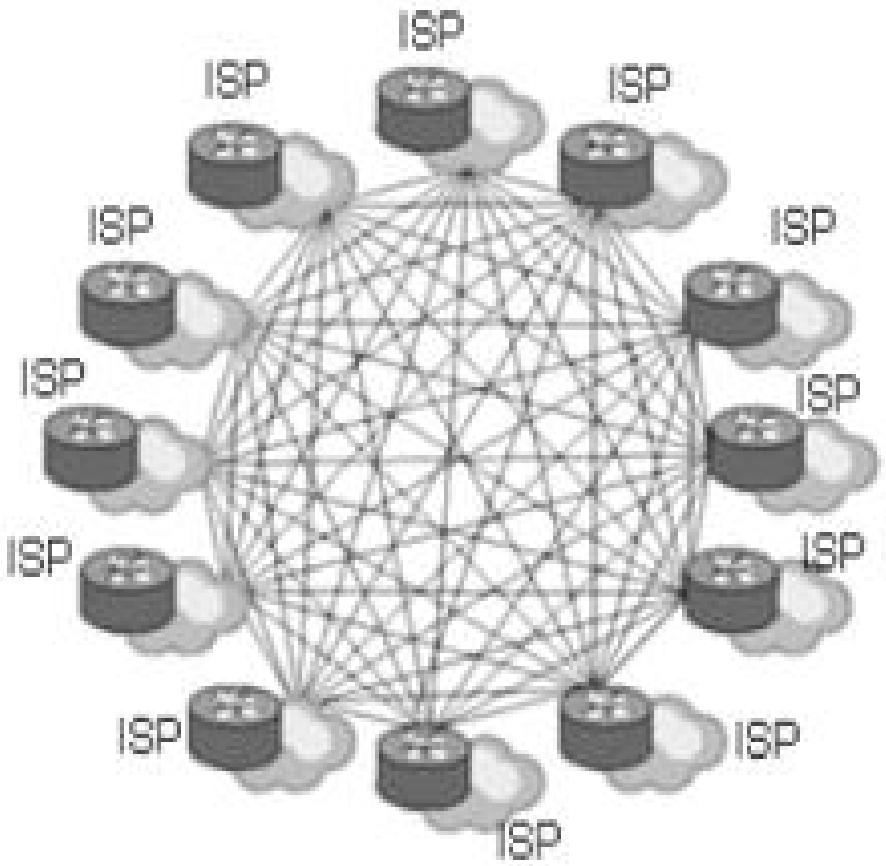
- 사용목적에 따라 확장성, 전송효율, 경제성 등을 고려하여 결정해야 함

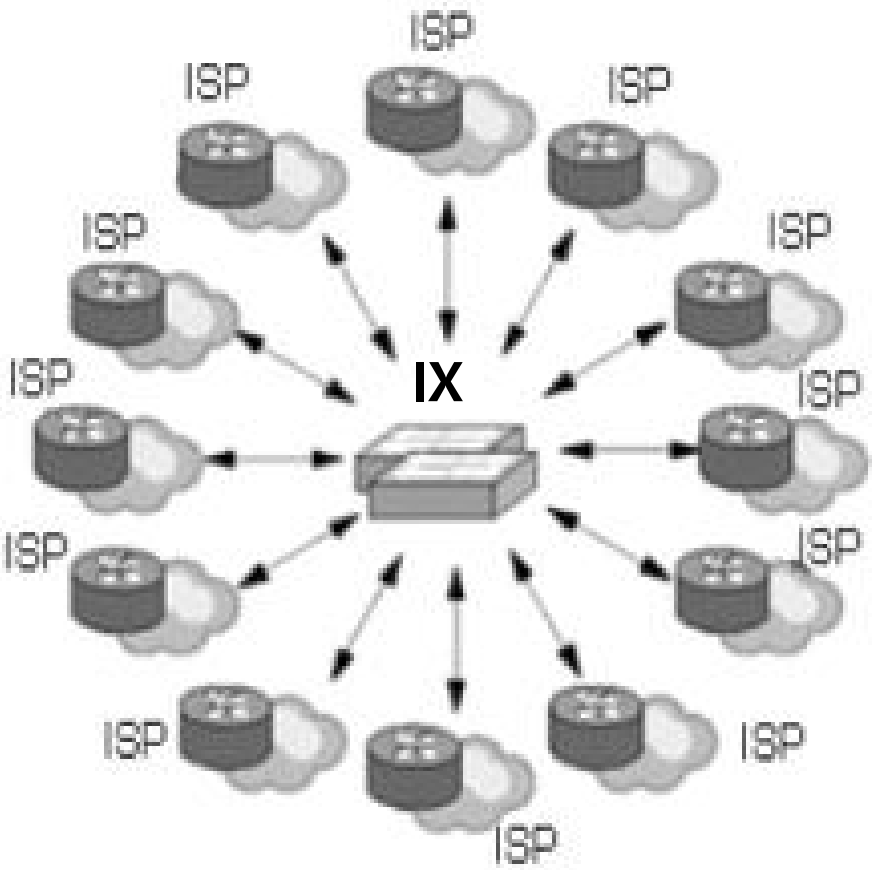
	Star	Bus	Tree	Ring	Mesh
케이블 수와 설치 비용	케이블이 적게 사용 중앙까지 연결이 필요	비교적 적게 듦 (경제적)	제어장치(hub)에 따라 다름	Star 형보다 저렴	N개의 노드에 대해 n(n-1)/2의 회선 필요 각 노드는 자신을 제외한 n-1개의 입출력 포트 필요
설치 비용	초기비용이 큼	경제적	초기 비용이 큼	설치 간단 제어절차 복잡	많은 회선으로 비교 부담
재구성	용이	어려움	제어 구조에 따라 다름	용이	어려움
장비추가	용이	용이	용이	어려움 (링 절단)	규모에 따라 다름

Internet & WAN



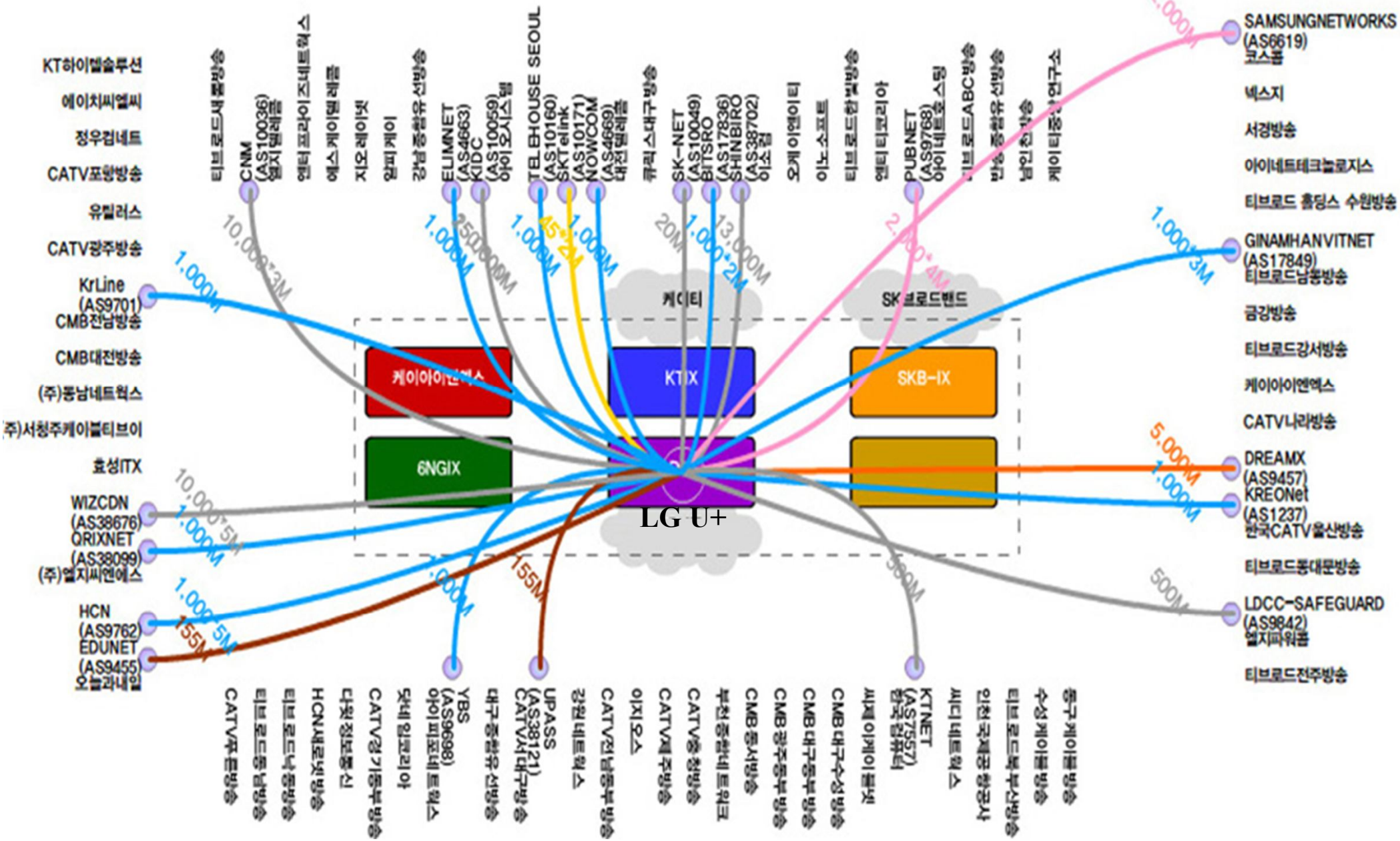
IX(Internet eXchange)





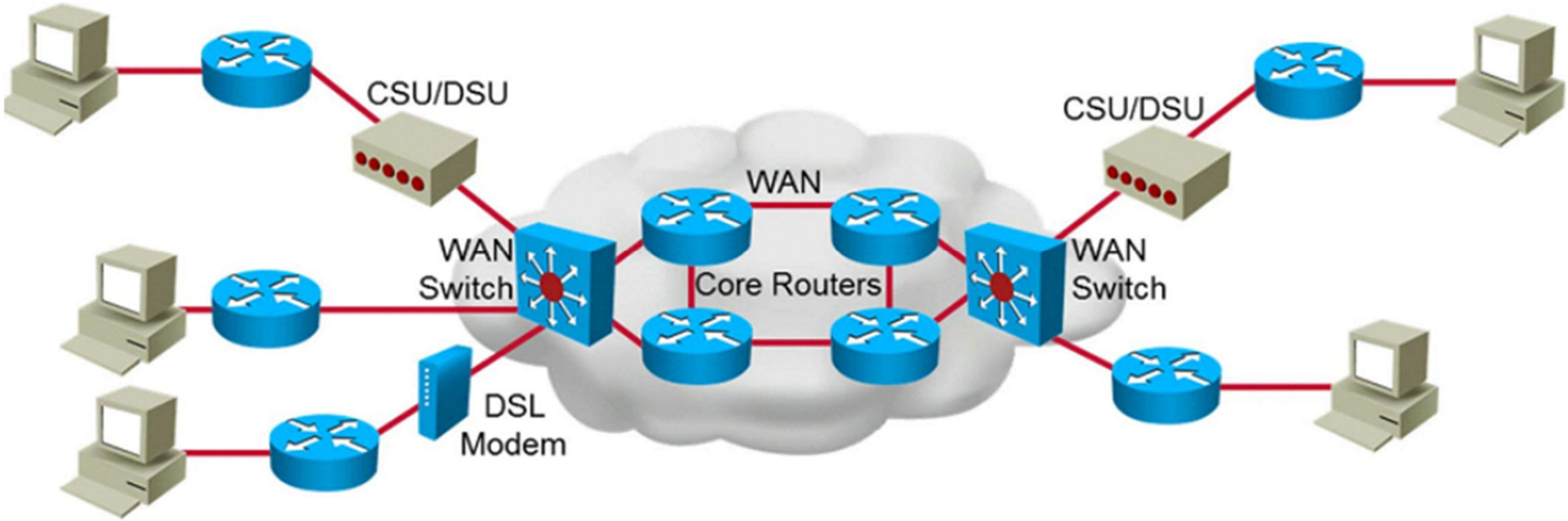
IX(Internet eXchange) –인터넷 트래픽을 상호 교환 시켜주는 역할 수행

Internet Connectivity Map Between ISP & IX

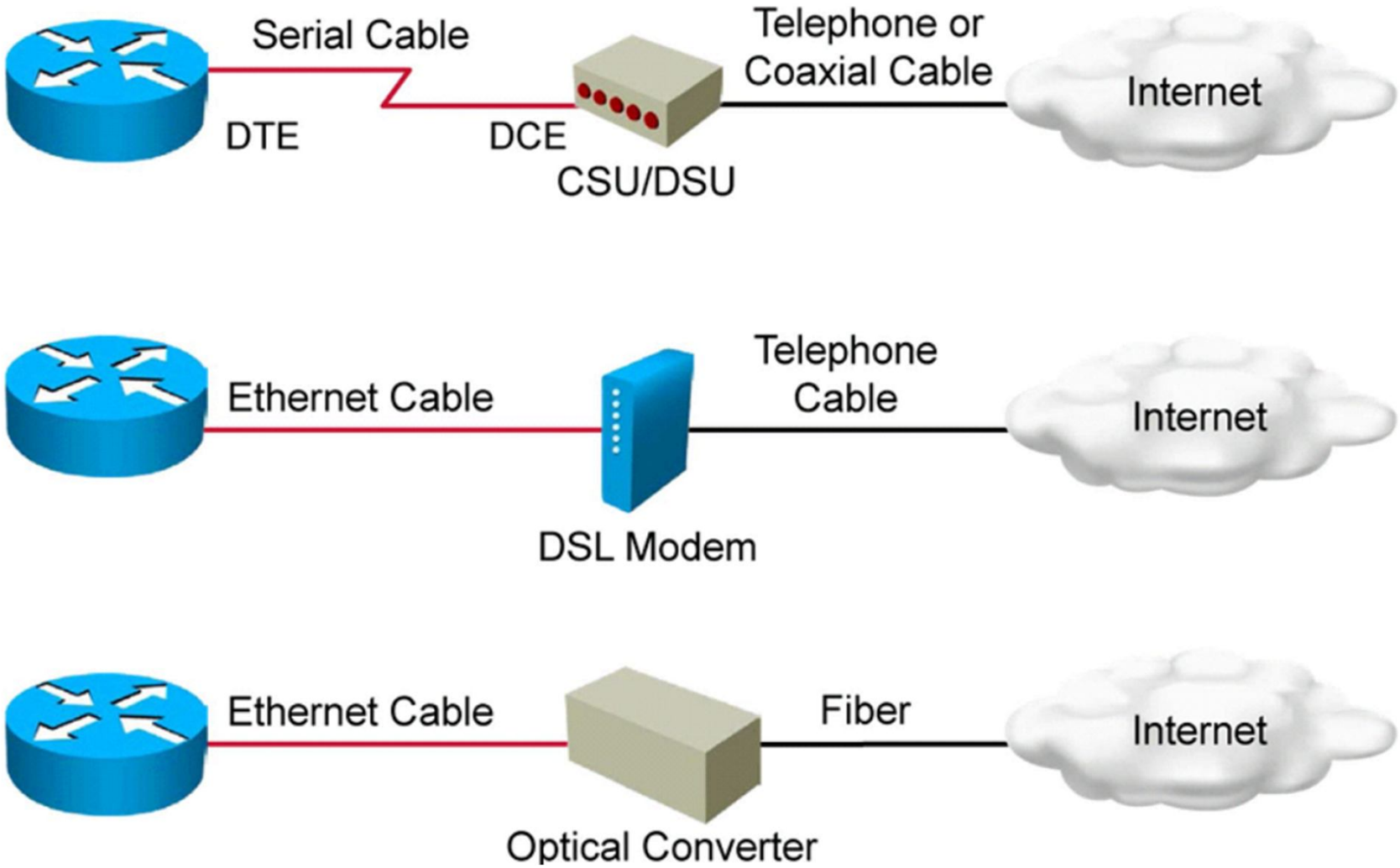


Internet & WAN Devices

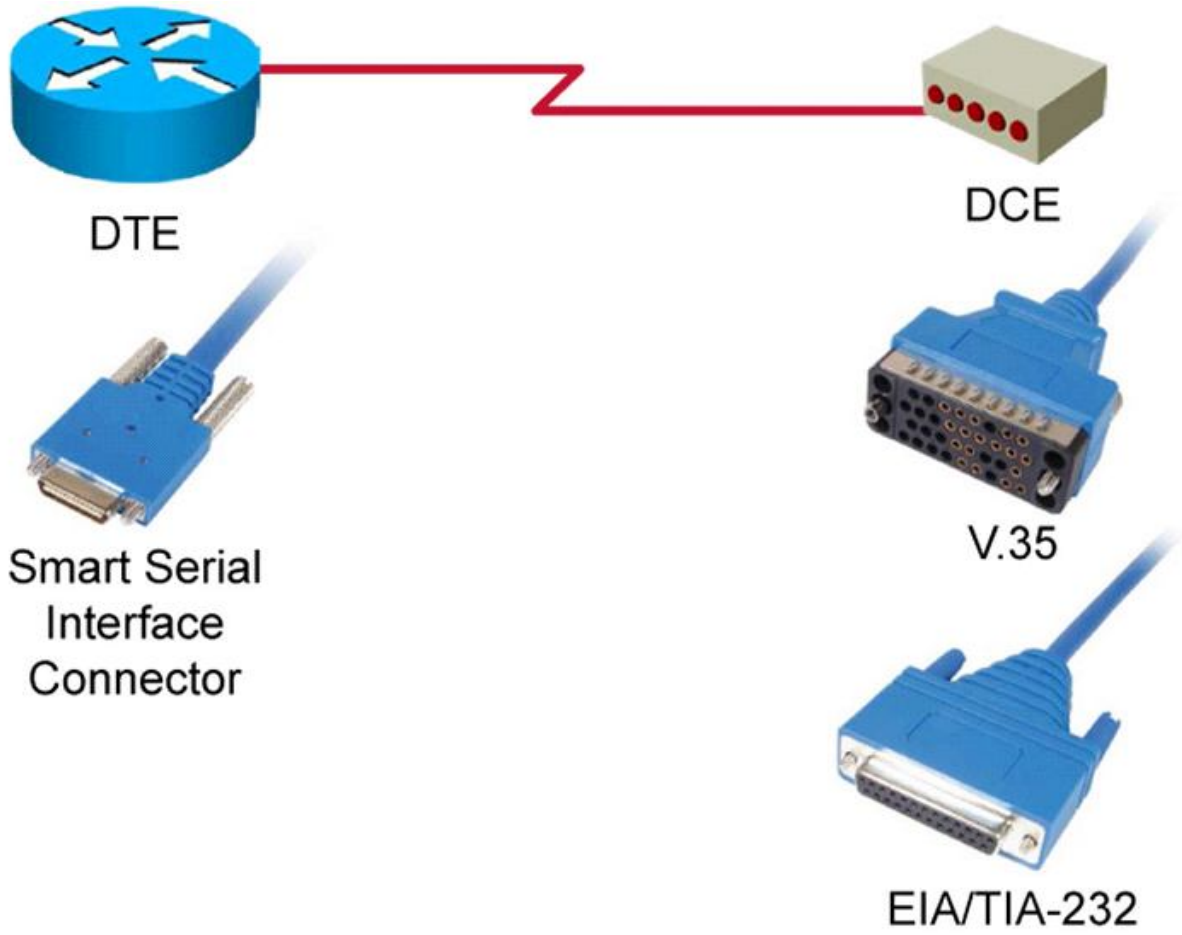
- Routers
- CSU/DSU
- WAN switches
- Core routers
- Modems



WAN Device & Cable



Serial Cable

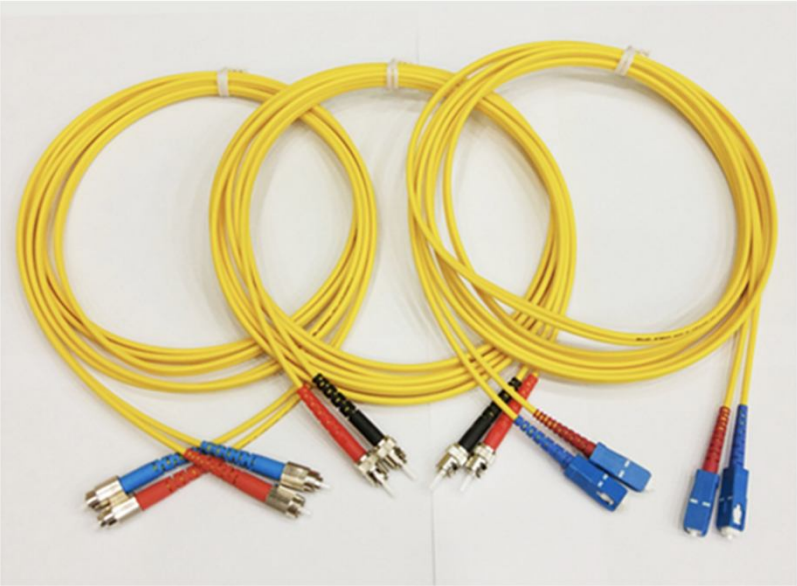


Serial Communication Links

- Typical WAN speeds for the U.S.:
  - T1 = (1.544 Mb/s)
  - T2 = 4 T1 lines (6 Mb/s)
  - T3 = 28 T1 lines (45 Mb/s)
  - T4 = 168 T1 lines (275 Mb/s)
- Typical WAN speeds for Europe:
  - E1 = (2 Mb/s)
  - E2 = 128 E0 lines (8 Mb/s)
  - E3 = 16 E1 lines (34 Mb/s)
  - E4 = 64 E1 lines (140 Mb/s)

Fiber Cable

- 데이터 전송을 위해서 광섬유로 만든 케이블
- 레이저를 이용해서 통신하기 때문에 장거리 & 고속 통신이 가능



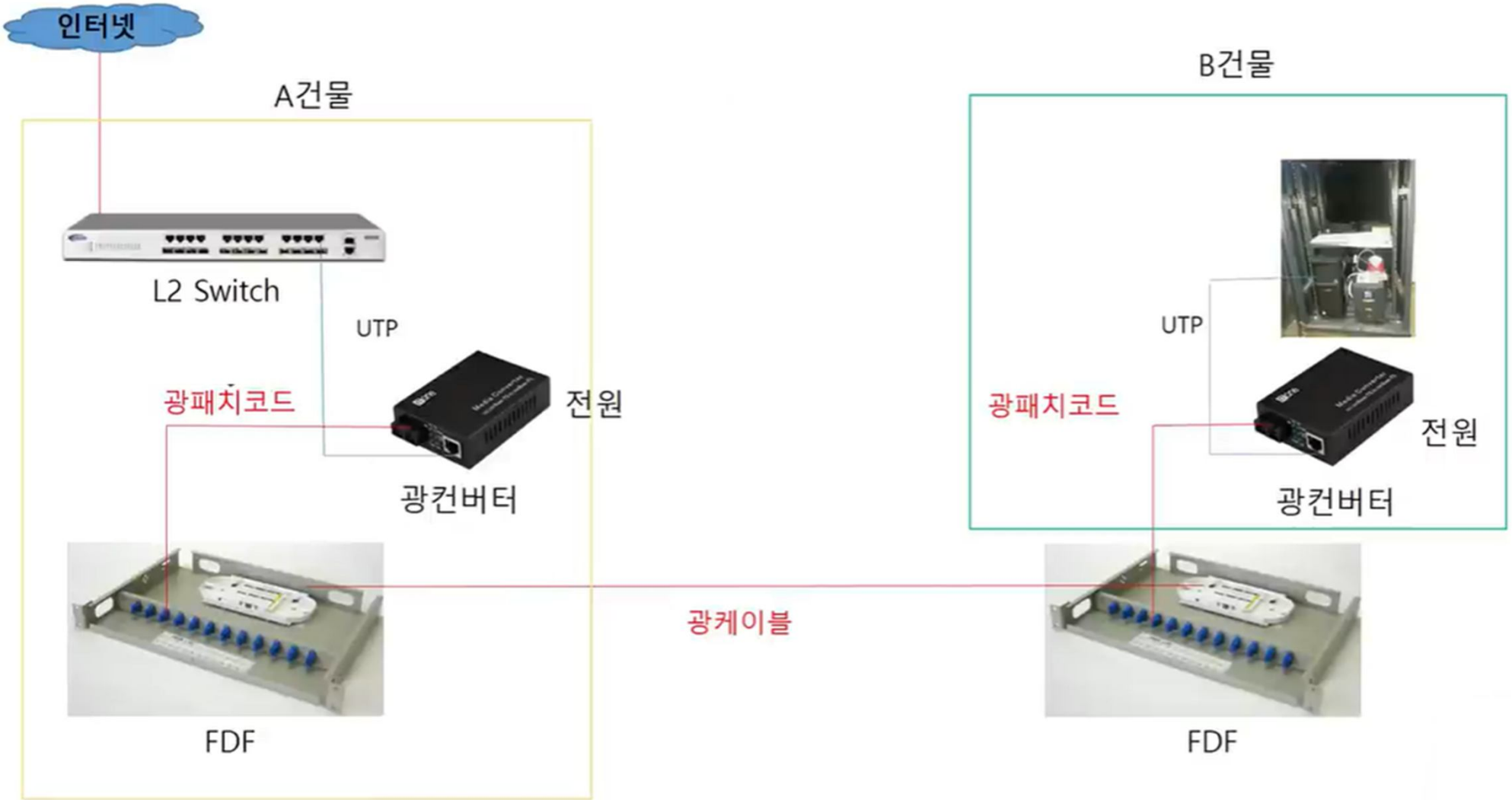
SM(Single Mode)광 케이블

- 장거리용(10~50Km)
- 장거리 고속 전송에 적합
- 가격이 MM광 케이블 보다 비쌈
- 노랑색 케이블

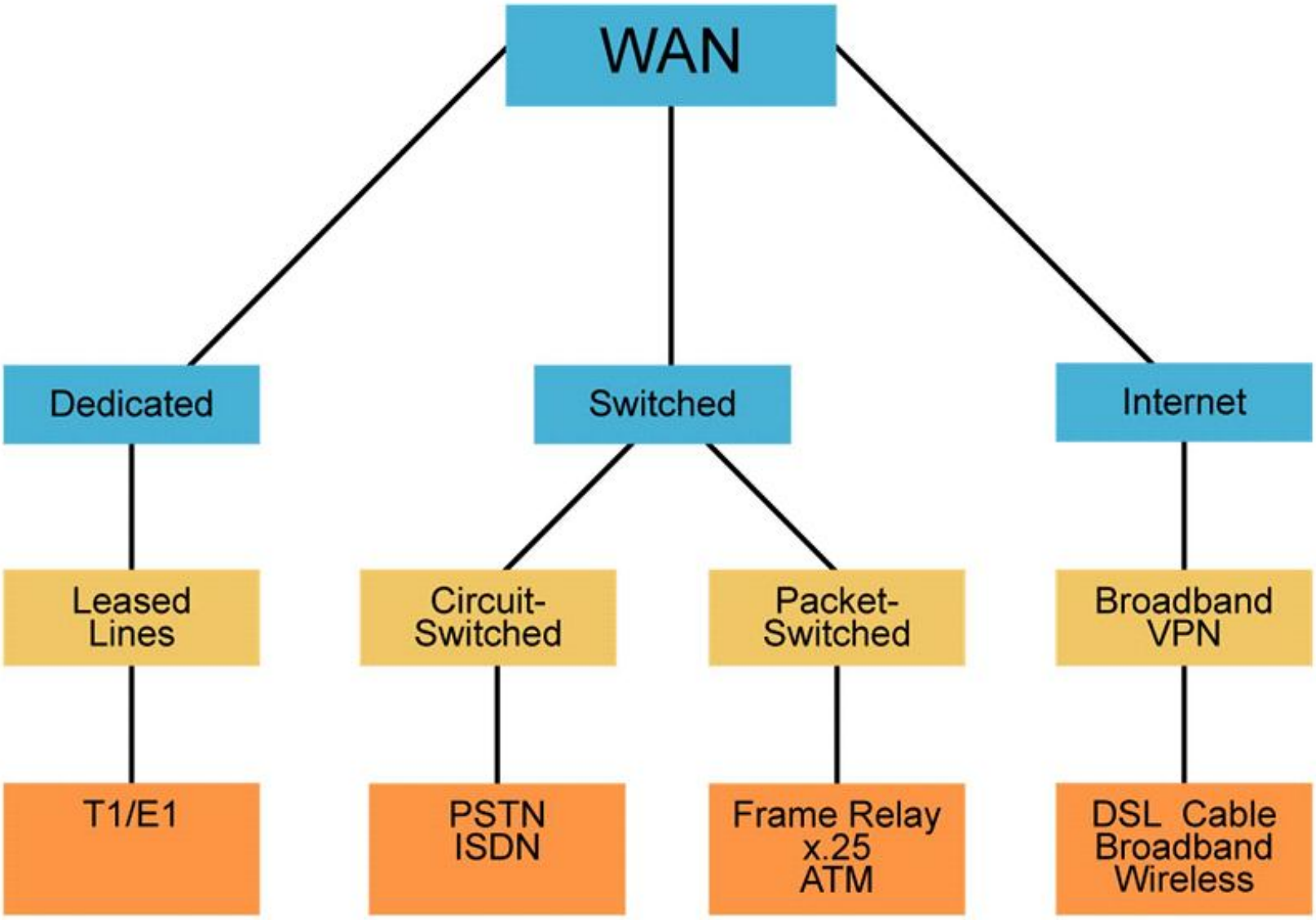
MM(Multi Mode)광 케이블

- 단거리용(1~10Km)
- 근거리 통신망에 적합
- 가격이 SM광 케이블 보다 저렴
- 주황색 케이블

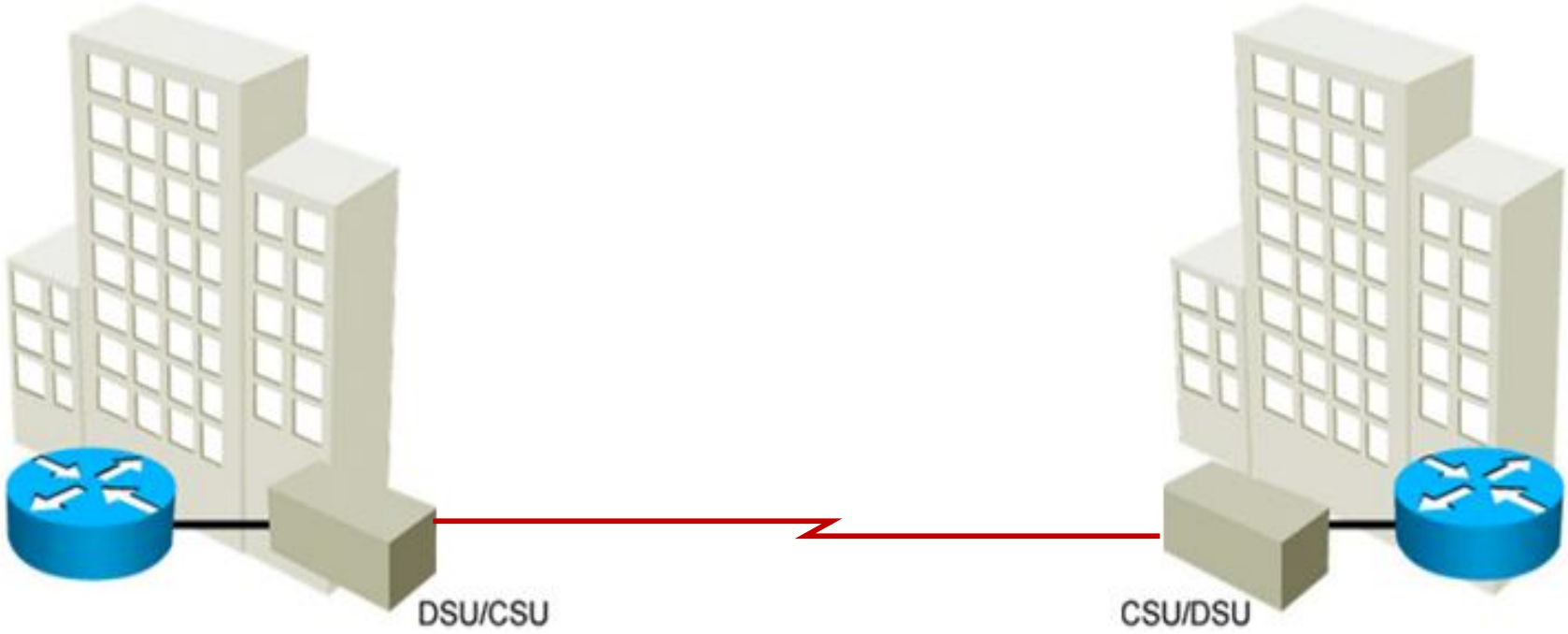
<<광케이블 커넥터 유형>>


인터넷
A건물
L2 Switch
UTP
광패치코드
전원
광컨버터
FDF
광케이블
B건물
UTP
광패치코드
전원
광컨버터
FDF

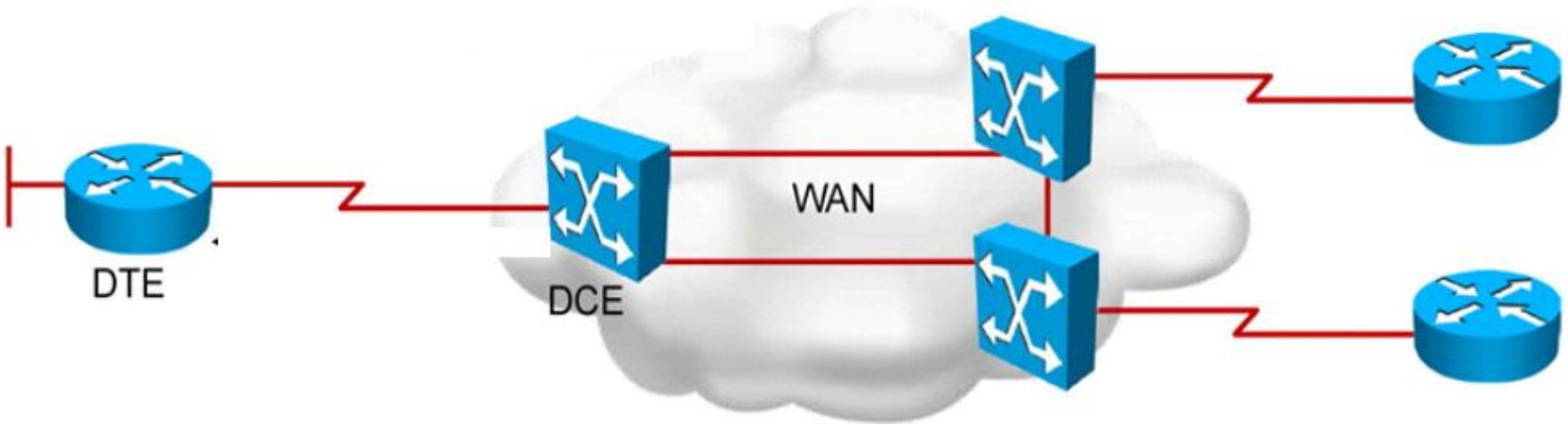
WAN(Wide Area Network) 구축 형태



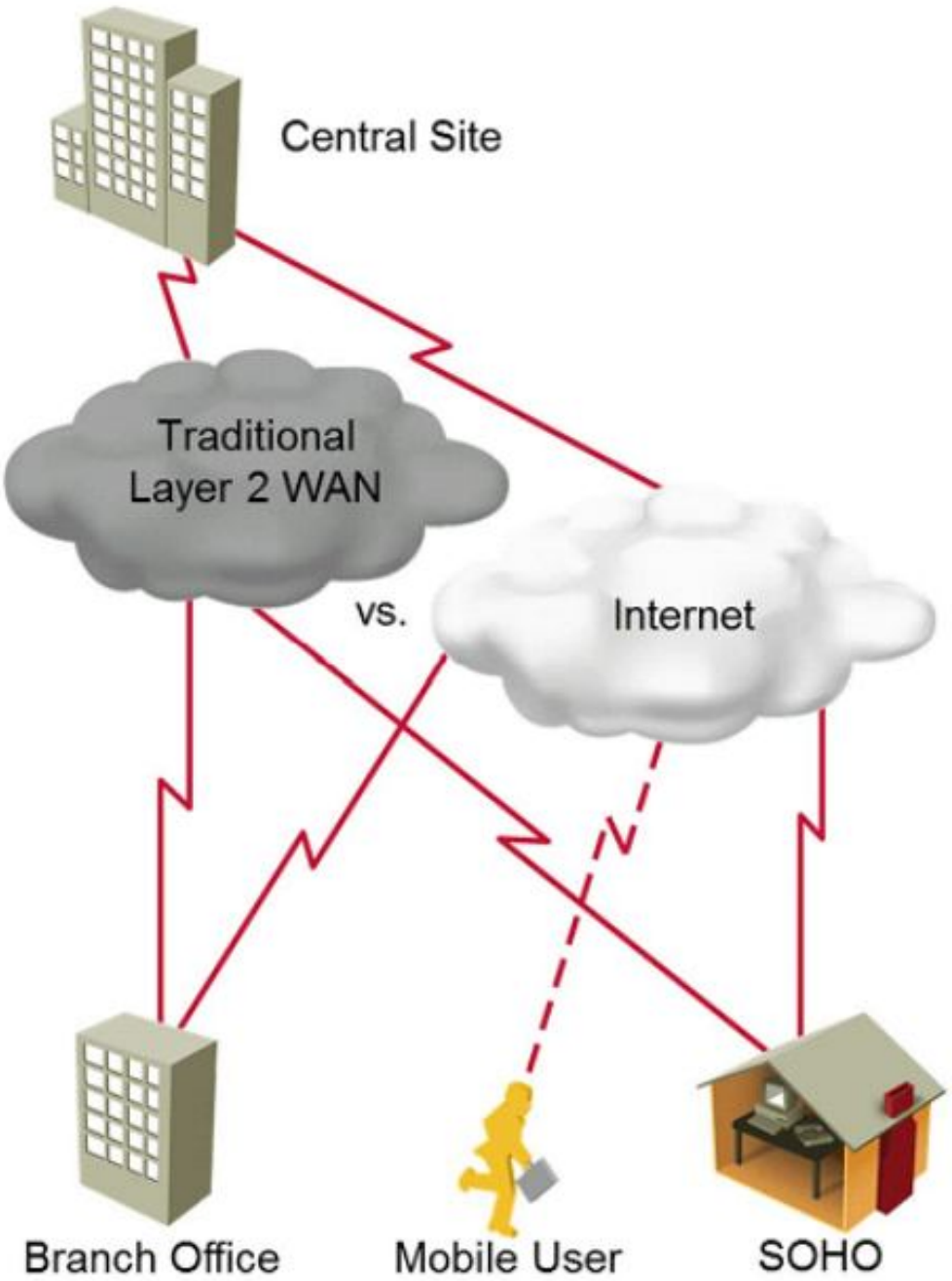
① Dedicated(Leased lines)



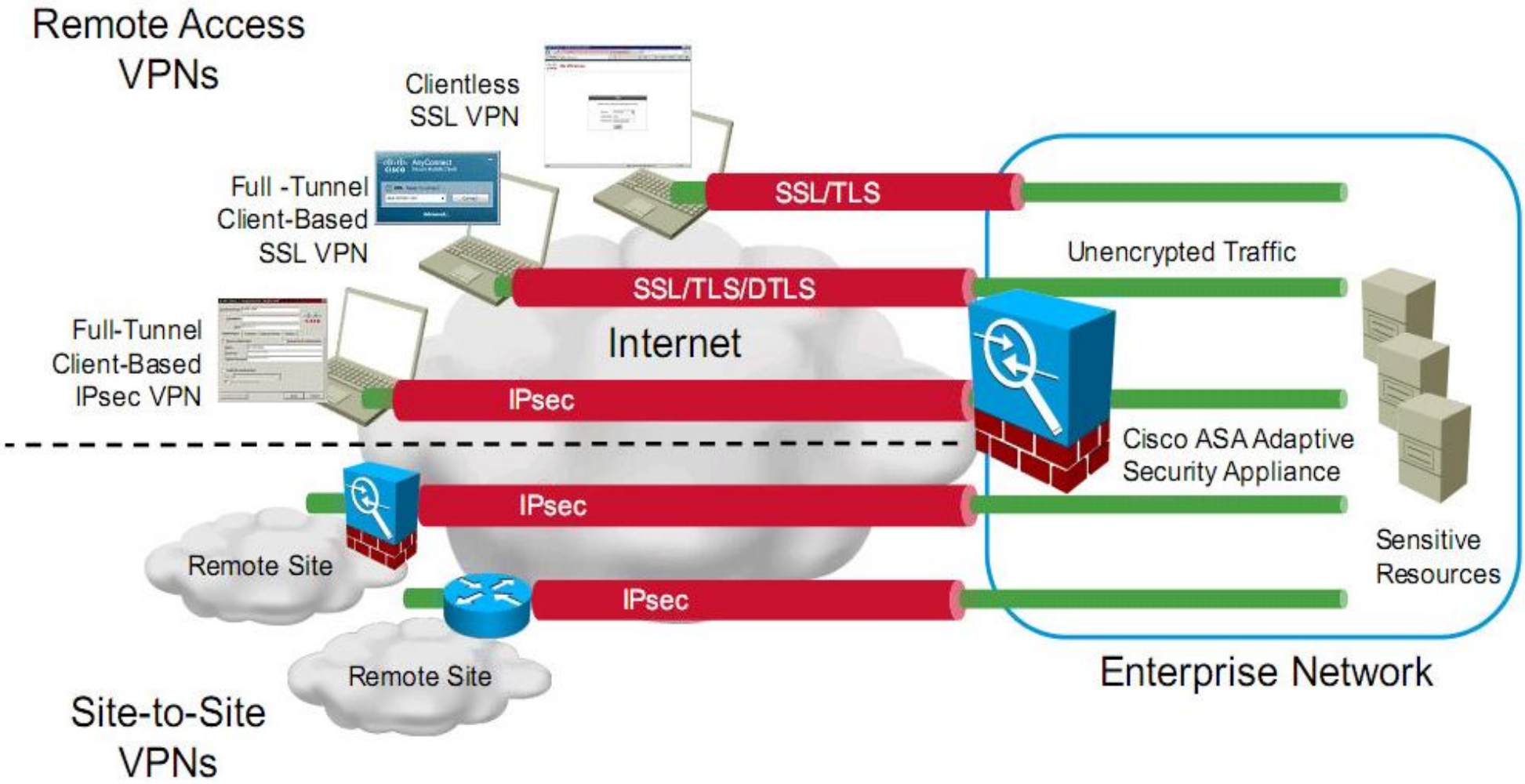
② Switched



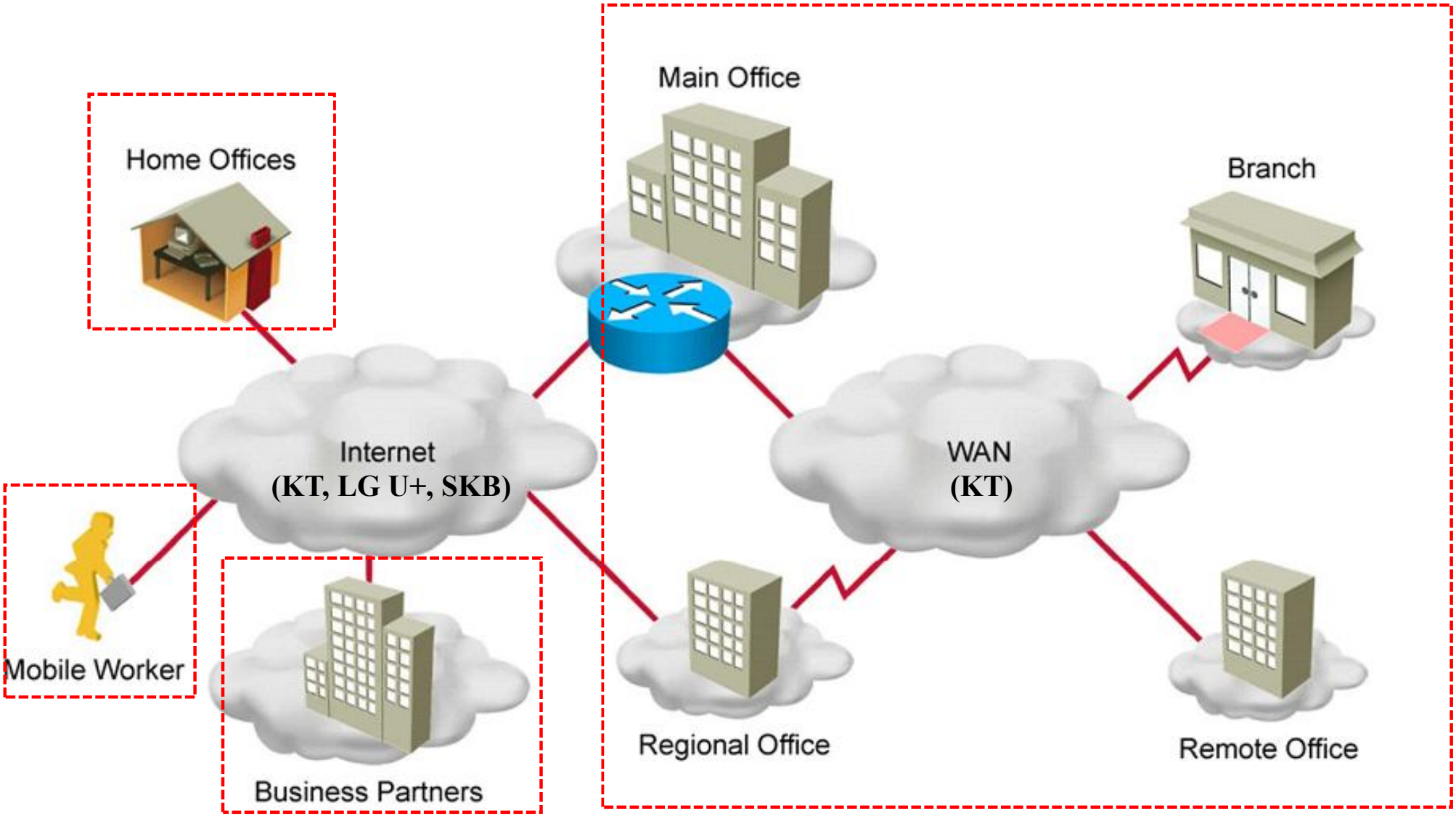
③ Internet


Central Site
Traditional
Layer 2 WAN
vs.
Internet
Branch Office
Mobile User
SOHO

Virtual Private Networks (VPNs)



Internet & WAN



WAN= A Collection of LANs